MILITARY SPECIFICATION

FUZE, POINT DETONATING, M505A2

PARTS FOR

1. SCOPE

1.1 This specification covers parts for the M505A2 fuze for use in 20-mm ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition

MIL-P-10025 - Packaging and Marking for Domestic Shipment of Inert Ammunition Components; General Specification For

STANDARDS

FEDERAL

Federal Test Method Standard No. 151 - Metals; Test Methods

MILITARY

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 - Inspection Terms and Definitions

DRAWINGS

ORDNANCE CORPS

D7259055 - Fuze, PD, M505A2 Assembly

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Sampling Tables and Procedures for Continuous Sampling by Attributes

IEL-D-7259055 MP - Index to List of Inspection Gages and Parts Gaged On: Fuze, PD, M505A2, Metal Parts

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with their applicable specifications.

3.2 Parts.-The parts shall comply with all requirements specified on applicable detail drawings referenced on Drawing D7259055, and with all requirements shown in applicable specifications.

3.3 Corrosion resistance.-The body, booster holder, cover, and safety spring retainer shall show no evidence of white salts or red rust when tested as specified in 4.5.1. Exposure time shall be as shown on the applicable drawings.

3.4 Spring displacement.-The body, spring and spring retainer assembly shall permit a 0.723 inch penetration of a test ram positioned against the safety spring when loaded with not less than 4.3 pounds nor more than 13.0 pounds. After release of the load, the safety spring shall not exceed maximum locational limit specified on the applicable component assembly drawing.

3.5 Spring Flexibility.-The safety spring arm shall be capable of withstanding a 120° displacement from the spring base without cracking or fracturing.

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3.6 Preproduction sample.-A preproduction sample, in the quantity specified in 4.3, shall be required prior to the start of regular production.

3.7 Workmanship.-All parts shall be free from chips, dirt, grease, corrosion, and other foreign material. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 The provisions of Specification MIL-G-2550 apply to this specification. Reference will be made to MIL-STD-109 for definition of inspection terms used herein.

4.3 Preproduction sample.-Prior to the start of regular production, the contractor shall submit a preproduction sample manufactured by the same methods and equipment that will be utilized in performing the work under the contract. The contracting officer will furnish instructions for shipment of the sample to a designated Government laboratory for examination and test.

4.3.1 The preproduction sample shall consist of seventy-five (75) sets of component parts and twenty-five (25) body, spring and spring retainer assemblies. The sample will be sent to and examined by a Government laboratory for compliance with the requirements of the applicable drawings and this specification.

4.3.2 Results of the examination and test will be furnished to the contractor. Approval or disapproval of the preproduction sample will be given by the responsible procuring activity. Instructions for the submission of new sample(s) in case of disapproval of the original sample will be issued by the procuring activity. Regular production shall not proceed until the preproduction sample has been approved.

4.4 Sampling

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4.4.1 Lot formation.-A lot shall consist of fuze metal parts and assemblies produced in accordance with the same drawings and drawing revisions, and same specification and specification revision (see 6.3).

4.4.2 Examination

4.4.2.1 Each unit of the lot, as applicable, shall be examined for the critical defects listed in 4.4.4. Any unit found to be defective shall be removed from the lot.

4.4.2.2 Lot sampling and acceptance criteria will be in accordance with MIL-STD-105 or ORD-M608-11, as appropriate. Judgement for lot acceptance will be based on either the individual defect characteristic or on classes of defect characteristics with individual defect characteristic limitations, as approved by the procuring activity prior to the initiation of production, with the following AQL's assigned (see 6.2):

4.4.2.2.1 Individual defect characteristic basis

. . . .	Major Defects	0.25%
	Minor Defects	0.40%

4.4.2.2.2 Defect characteristic class basis

Major Defects	1.50%
Minor Defects	2.50%

Individual defect characteristic AQL's associated with the class basis, shall be

Major Defects	0.40%
Minor Defects	0.65%

4.4.3 Test inspection.-The tests indicated below will be performed in accordance with the methods specified in 4.5. Acceptance criteria are as specified below:

TABLE I

Test	Part
Corrosion resistance	Body
	Booster holder
	Cover
	Safety spring retainer
Hardness	Cover
	Rotor
Spring flexibility	Safety spring

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4.4.3.1 Corrosion resistance.-A double sampling plan using the method of MIL-STD-105 will be applied to the test for corrosion resistance. Sample sizes will be twenty-five (25) and fifty (50) of each of the parts specified in Table 1 with acceptance criteria of zero (0) and two (2), respectively, for each of the parts. Any part which fails to meet the requirement of 3.3 will be classed defective and penalty to the lot will be assigned according to the acceptance criteria.

4.4.3.2 Spring flexibility.-A single sampling plan using the method of MIL-STD-105 will be applied to the test of spring flexibility. The sample size for this test will be 25 springs and the acceptance number will be zero (0). Any spring which shows evidence of a crack or fracture will be classed defective causing the lot to be finally rejected.

4.4.3.3 Hardness.-Double sampling plans using the method of MIL-STD-105 will be applied to the tests for hardness. Sample sizes will be twenty-five (25) and fifty (50) fuze covers, respectively, with acceptance criteria corresponding to 1.0% AQL's

4.4.4 Classification of defects

4.4.4.1 Booster holder (see Dwg. C7258826 covering a detail of Dwg. 7259055.)

Categories and defects	Method of Inspection
Critical	
None assigned	
Major	
101. Length from forward edge of flange to top of .37700 ball	Gage
102. Crimping groove missing	Visual
103. Major diameter of thread	Gage
104. Pitch diameter of thread	Gage
105. Concentricity of outside diameter of body with pitch diameter of thread	Gage
106. Concentricity of flange diameter with pitch diameter of thread	Gage
107. Concentricity of .37700 ball with pitch diameter of thread	Gage
108. Body diameter forward of thread, max	Gage
109. Length from forward edge of flange to end of thread, max	Gage
110. Length of thread, min	Gage
111. Finish of spherical radius	Measuring device
112. Foreign matter	Visual (see 4.4.5.2)
113. Damage	Visual

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Minor

201. Diameter of flash hole----------Gage
202. Diameter of booster cavity, min---------Gage
203. Depth of booster cavity from forward edge of flange----------Gage
204. Diameter of flange, max ----------Gage
205. Protective coating damaged or missing---Visual
206. Burr----------Visual
207. Finish except spherical radius----------Visual - Feel
208. Diameter of undercut, rear of thread----Gage
209. Height, crimping lip, min----------Gage

4.4.4.2 Rotor (see Dwg. C7259058, covering a detail of Dwg. D7259055)

Categories and defects	Method of inspection
Critical	
None assigned	
Major	
101. Diameter of detonator cavity, min-------Gage	
102. Depth of detonator cavity----------Gage	
103. Location of large flat----------Gage	
104. Web thickness ----------Gage	
105. Burr above spherical surface----------Visual	
106. Foreign matter----------Visual	(see 4.4.5.2)
Minor	
201. Outside diameter----------Gage	
202. Diameter of flash hole, min.----------Gage	
203. Concentricity of detonator cavity with outside diameter ----------Gage	
204. Concentricity of flash hole with detonator cavity----------Gage	
205. Finish ----------Visual - Feel	(See 4.4.5.3)
206. Burr, except above spherical surface----Visual	

4.4.4.3 Spring, rotor safety (see Dwg. D7258824, covering a detail of Dwg. D7259055

Categories and defects	Method of inspection
Critical	
None assigned	

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Major

101. Profile of spring arm from base to tip--Gage
102. Protective coating damaged or missing---Visual
103. Foreign matter----------Visual (see 4.4.5.2)

Minor

201. Burrs----------Visual

4.4.4.4 Cover (see Dwg. C7258821, covering a detail of Dwg. D7259055)

Categories and defects	Method of inspection

Critical

None assigned

Major

101. Inside diameter at large end----------Gage
102. Length from base to .300 datum diameter-Gage
103. Outside contour of tapered portion------Gage
104. Length from inside of nose to .300 datum diameter ----------Gage
105. Protective coating or cladding damaged or missing ----------Visual
106. Cracked, split, or deformed----------Visual
107. Wall thickness, min.----------Gage
108. Foreign matter ----------Visual (see 4.4.5.2)

Minor

201. Wall thickness, max----------Gage
202. Nose thickness----------Gage
203. Burrs----------Visual

4.4.4.5 Body (see Dwg. D7258820, covering a detail of Dwg. D7259055)

Categories and defects	Method of inspection

Critical

1. Minimum groove profile ----------Gage

Major

101. Major diameter of external thread-------Gage
102. Pitch diameter of external thread-------Gage
103. Pitch diameter of internal thread-------Gage
104. Minor diameter of internal thread-------Gage
105. Concentricity of booster holder cavity with pitch diameter of internal thread-Gage
106. Diameter of spring groove, min----------Gage
107. Length from nose to rear of cover crimping shoulder----------Gage
108. Length from base to bottom of .37700 ball-Gage

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109. Length from spring groove to rotor safety spring retainer shoulder ----------Gage
110. Finish of rotor seat----------Measuring device
111. Protective coating damaged or missing---Visual
112. Shoulder diameter (after knurling)------Gage
113. Knurl missing or incomplete ----------Visual
114. Profile of tapered surfaces from nose to crimping shoulder----------Gage
115. Chamfer at entrance to firing pin cavity ----------Measuring device
116. Length of internal thread, min (effective thread)----------Gage
117. Foreign matter in rotor seat or spring slot----------Visual (see 4.4.5.2)
118. Burr----------Visual
119. Spring slot width, min----------Gage
120. Length of spring slot, min----------Gage
121. Location of spring slot from shoulder, min ----------Gage
122. Concentricity of firing pin hole with pitch diameter of internal thread-----Gage
123. Finish of firing pin hole----------Measuring device
124. Diameter, cylindrical surface, nose end-Gage

Minor

201. Length of thread relief at shoulder, max ----------Gage
202. Length of thread relief at base---------Gage
203. Total length ----------Gage
204. Finish, except firing pin hole and rotor seat----------Visual - feel
205. Radius at rear of knurled shoulder, max-Gage
206. Depth of counterbore from base to start of internal thread----------Gage
207. Length of internal thread including counterbore and runout ----------Gage
208. Diameter of booster holder cavity-------Gage
209. Centrality of spring slot with pitch diameter of external thread----------Gage
210. Wall thickness variation at nose--------Gage

4.4.4.6 Body, spring, and spring retainer assembly (see Dwg. D7259056 covering a detail of Dwg D7259055)

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Categories and defects	Method of inspection
Critical	
1. Location of spring tip, more than .002 above .080 maximum dimension	Gage
2. Spring displacement	Measuring device (see 4.4.5.4)
Major	
101. Location of spring tip, .080 max	Gage
102. Location of spring tip, .090 min (with minimum rotor in place)	Gage
103. Loose or missing rotor safety spring retainer	Manual-Visual (see 4.4.5.1)
104. Gap between body shoulder and rotor safety spring retainer	Gage
105. Diameter of firing pin hole	Gage
106. Rotor safety spring retainer protruding above fuze body taper	Gage
107. Burr	Visual
108. Crack in rotor safety spring retainer	Visual
109. Foreign matter	Visual (see 4.4.5.2)

4.4.4.7 Firing pin (see C7258823, covering a detail of Dwg D7259055

Categories and defects	Method of inspection
Critical	
1. Flange thickness, min	Gage
2. Flange missing	Visual
Major	
101. Diameter of flange	Gage
102. Large body diameter	Gage
103. Profile of point	Gage
104. Arming spring locking shoulder missing or improper	Visual
Minor	
201. Finish	Visual - Feel
202. Body length from under side of flange, min.	Gage

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4.4.4.8 Sealed packing container

Categories and defects	Method of inspection
Critical	
None assigned	
Major	
None assigned	
Minor	
201. Container damaged	Visual
202. Marking incorrect, incomplete, illegible or missing	Visual

4.4.5 Defect standards

4.4.5.1 Any rotor safety spring retainer that moves when subjected to finger pressure (applied by twisting the retainer between the thumb and index finger), or that moves when a probe is applied against the side of the retainer at the spring slot will be classed defective.

4.4.5.2 Any part or subassembly showing the presence of dirt, oil, grease, corrosion, or other foreign substance will be classed defective.

4.4.5.3 Should conformance of outside diameter or detonator cavity wall to requirement be doubtful, the measuring devices prescribed on the pertinent IEL shall be used.

4.4.5.4 Any assembly that fails to meet the requirement of 3.4 will be classed defective. Assemblies may be tested without safety spring retainer in place.

4.4.6 Inspection Equipment

4.4.6.1 Government verification inspection will be accomplished with the use of gages and equipment specified in "Index to List of Inspection Gages and Parts Gaged on: Fuze, PD, M505A1, Metal Parts," GL7259308.

4.5 Test methods

4.5.1 Corrosion resistance.-The samples selected for the test will be tested in accordance with Method 811 of Federal Test Method Standard No. 151.

4.5.2 Spring flexibility.-The rotor safety spring will be tested by displacing the spring arm 120° from the spring base using resting device shown on Dwg. D7259500. The sample springs tested will not be returned to the lot.

4.5.3 Hardness.-The sample selected will be tested in accordance with method 243 of Federal Test Method Standard No. 151, utilizing appropriate Rockwell Hardness Testers and prescribed scales.

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4.5.3.1 Cover.- The sample will be sectioned axially and one hardness impression will be taken on the inside surface at approximately the center of each section. The average hardness of the two sections shall be the sample hardness. Cladding will be removed from the areas to be tested by a process which will not alter the hardness of the basis metal.

4.5.3.2 Rotor.-The samples will be prepared and tested as prescribed on Dwg. C7259058, covering a detail of Dwg. D7259055. The average hardness of the four impressions shall be the sample hardness.

5. PREPARATION FOR DELIVERY

5.1 Packaging - Level C.-Packaging shall be in accordance with Specification MIL-P-10025.

5.2 Packing - Level C.-Packing shall be in accordance with Specification MIL-P-10025

5.3 Marking.-Marking on packing containers shall be in accordance with Specification MIL-P-10025.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 AQL's.-The optional use of AQL values for either individual defects or classes of defects, as approved by the procuring activity prior to the initiation of production, is authorized to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the process.

6.3 Application.-This specification may apply to a component as well as to sets of fuze metal parts and assemblies.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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